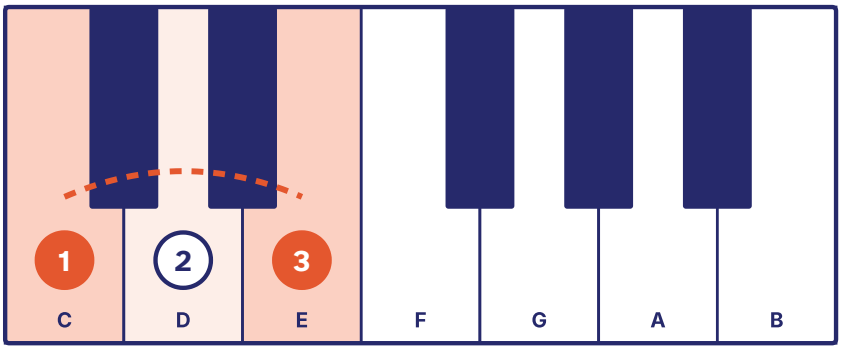
PART I · PITCH · 6

Counting intervals

How far apart are two notes? That distance is an **interval**. To give it a number, **count the notes**: the first note is 1, and you count every letter up to the second, both ends included.

THE SECRET

Count the notes, both ends included: C D E is 3 notes, a 3rd.

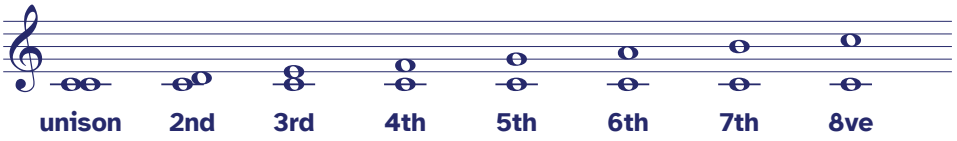


C to E: count the notes, 1 2 3: a 3rd

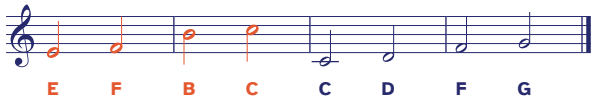
THE RULE

Count the notes for the number: start at 1 on the first note. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

Interval numbers from C



The smallest steps: semitones and tones



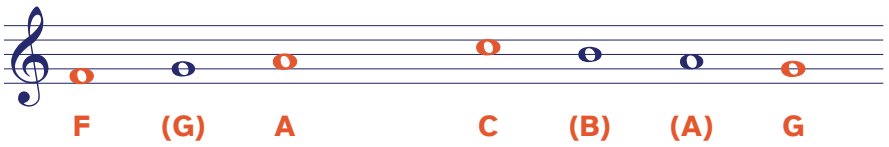
E–F and B–C: no black key between, a semitone. C–D and F–G: a black key between, a tone.

From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

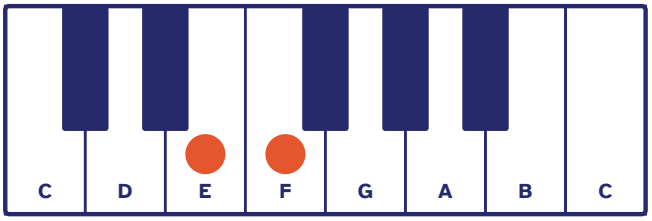


1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

EXAMPLE 2

Tone or semitone?

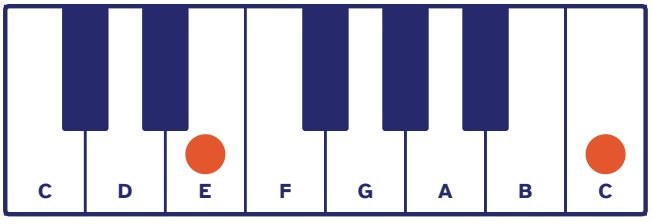


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

EXAMPLE 3

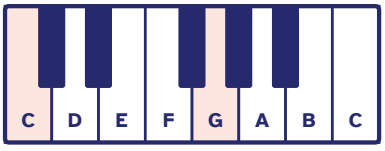
Name an interval on the keyboard



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

TRY IT · 10 SECONDS



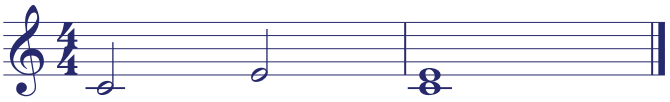
Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

Melodic or harmonic?



1 **The first bar:** C, then E, one after the other.

2 **The second bar:** C and E at the same time.

3 **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

GAME

Count the notes
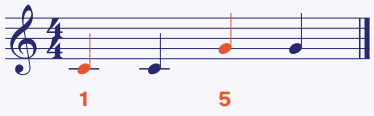
1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every note, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of any song, then count the notes between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: C to E is 3 notes, not 2 gaps.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 42 · Major scales p. 90

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

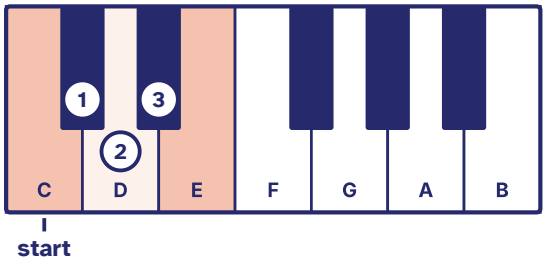
PART I · PITCH · 7

Major, minor and perfect

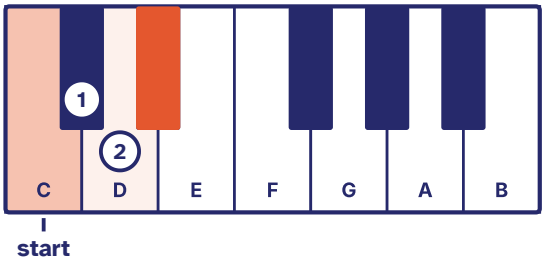
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the keys in between**, black and white: the keys you do not play.

THE SECRET

Count the keys between: 3 make a major 3rd, 2 a minor 3rd.



C to **E**: 3 keys between. A **major** 3rd: bright.



C to **E♭**: 2 keys between, one fewer. A **minor** 3rd: dark.

One key fewer between: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the keys in between for the size (major or minor): black and white, not the two you play.

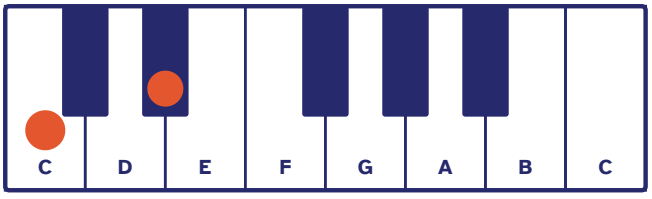
NUMBER	KEYS BETWEEN: SMALLER	MAIN SIZE
3rd	2 (minor)	3 (major)
4th		4 (perfect)
5th		6 (perfect)
6th	7 (minor)	8 (major)
octave		11 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one key fewer between.

4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

EXAMPLE 1

Major or minor 3rd?



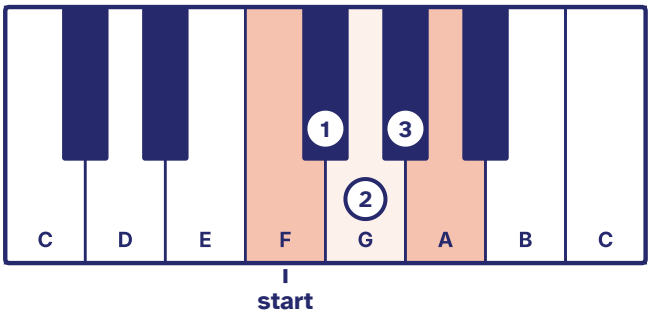
1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the keys in between** C and the red black key, E♭: C♯, D …
3. **A major 3rd has 3 between.** How many here?

C to E♭ has **2 keys between**: a **minor 3rd**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 2

Name F up to A

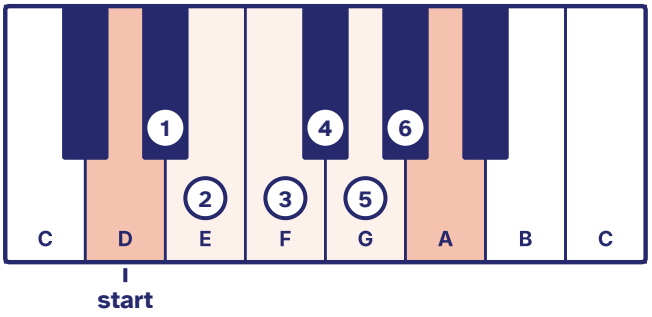


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the keys in between:** they are numbered.
3. **3 between, or 2?**

F to A has 3 keys between: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A



1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the keys in between: the numbered keys.
3. **A 5th with 6 between** is its main size. What is that called?

D up to A is a **perfect 5th**: 6 keys between.

GAME

Perfect, major or minor?

1	2	3	4
C–G	D–F	C–E	E–A

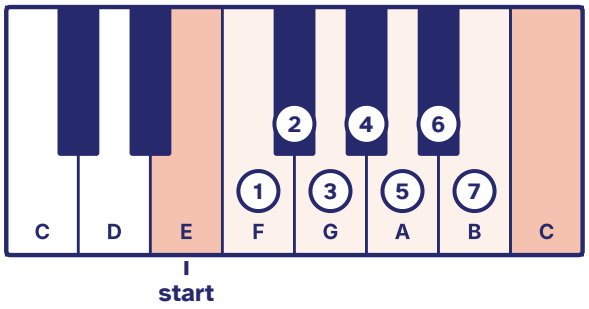
Count the notes for the number, then the keys in between for the size.

Answer: 1 perfect 5th (6 between) · 2 minor 3rd (2) · 3 major 3rd (3) · 4 perfect 4th (4).

PART I · PITCH · 7 · CONTINUED

EXAMPLE 4

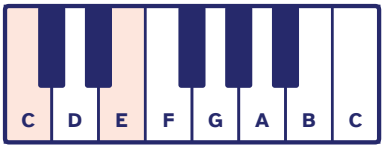
Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the keys in between.
3. **A major 6th** has 8 between. Is this one more or one fewer?

E up to C has 7 keys between, one fewer than major: a **minor 6th**.

TRY IT · 10 SECONDS
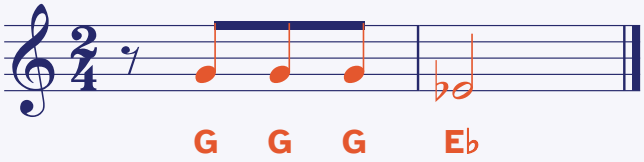


Play C–E, then C–E♭.

Bright, then dark. One key makes the difference: the major 3rd has three keys between, the minor 3rd two.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to **E♭** is three notes, a 3rd, with 3 keys between (F♯, F, E): a **major 3rd**.

Try it. Count the keys between G and E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every key between, black and white.
- × Counting the two notes you play: count only the keys in between.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO
Counting intervals p. 39 · Chords and triads p. 107

PRACTISE IT
Workbook Pitch 4 · Online: Library › Intervals